

Predicting Song Popularity and Enhancing Spotify's Recommender Systems: A Business Data Analytics Approach

Team Members,

- 1. Anjelina Sajan**
- 2. Kirthan Shekhar**
- 3. Lakshmy Prakash**
- 4. Saya Santhosh**

PROJECT OVERVIEW

Business Context:

Spotify needs to predict song popularity to enhance recommendations.

Dataset spans 1998-2020 U.S. Spotify charts

Goal: Improve user engagement through better song predictions

Key Objectives:

1. Develop three predictive models for song popularity
2. Optimize model selection based on financial metrics
3. Analyze feature patterns for improved recommendations

Dataset Scope:

27 features including audio characteristics

1500+ songs across multiple genres

Popularity scores as target variable

Methodology:

Data preprocessing and feature engineering

Model development and comparison

Profit-based optimization

Clustering analysis for insight generation

PROJECT PHASES

Phase 1: Predict Song Popularity (Q1)

- Develop and evaluate predictive models.
- Identify the best-performing model for predicting popularity.

Phase 2: Profit Maximization (Q2)

- Incorporate cost-benefit analysis into model selection.
- Optimize predictions to maximize revenue and minimize errors.

Phase 3: Feature Insights and Clustering (Q3)

- Analyze the role of song characteristics (e.g., valence) in predicting success.
- Cluster songs by danceability and energy to identify patterns.

Phase 1: Predict Song Popularity (Q1)

The problem is that Spotify wants to improve its song recommendation system by predicting which new songs are likely to become popular. Specifically, they want to predict whether a song's popularity score will exceed 50 based on various song characteristics like danceability, energy, tempo, etc.

The goal is to develop and compare three different predictive models, using provided historical data from 1998-2020, and select the best model based on accuracy metrics to help Spotify prioritize songs that are more likely to engage users effectively.

"Key features like danceability, energy, tempo, and valence are analyzed to determine their relationship with popularity scores."

OBJECTIVES

- Develop and compare three different predictive models to forecast whether a song's popularity score will exceed 50, using appropriate data preprocessing and feature selection techniques.
- Create an appropriate data preprocessing pipeline including handling missing values, encoding categorical variables, and selecting relevant features for prediction.
- Identify the best performing model based on accuracy metrics and justify why it's the most suitable choice for predicting song popularity.
- Provide actionable managerial insights for Spotify about which song attributes are associated with higher engagement and popularity, to help them optimize playlists and song recommendations.

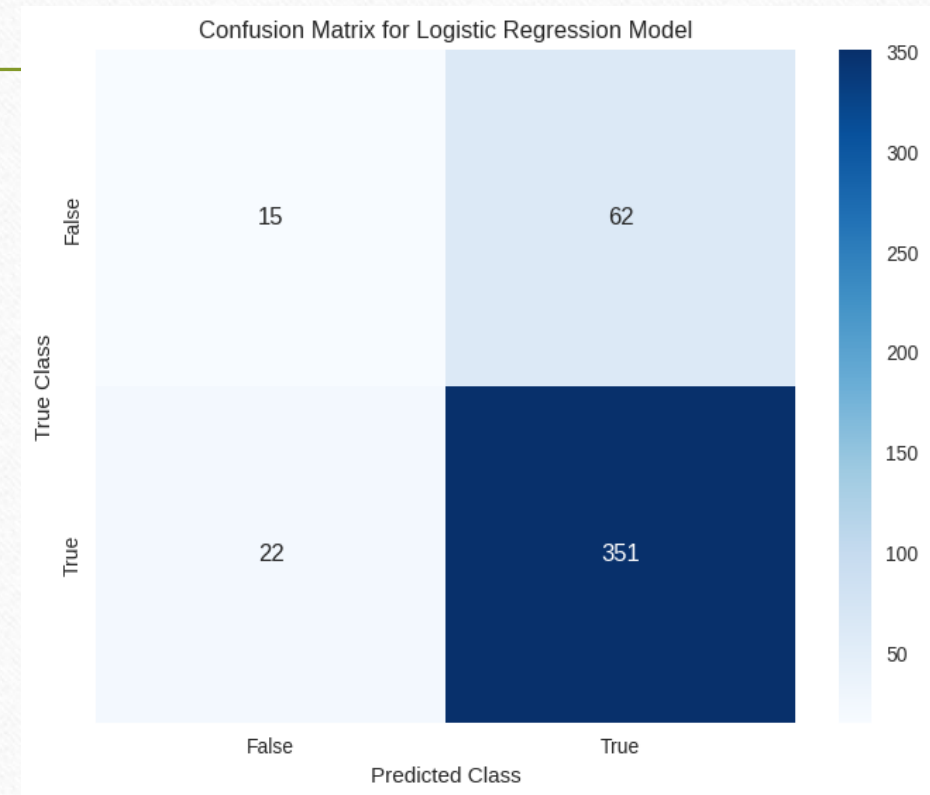
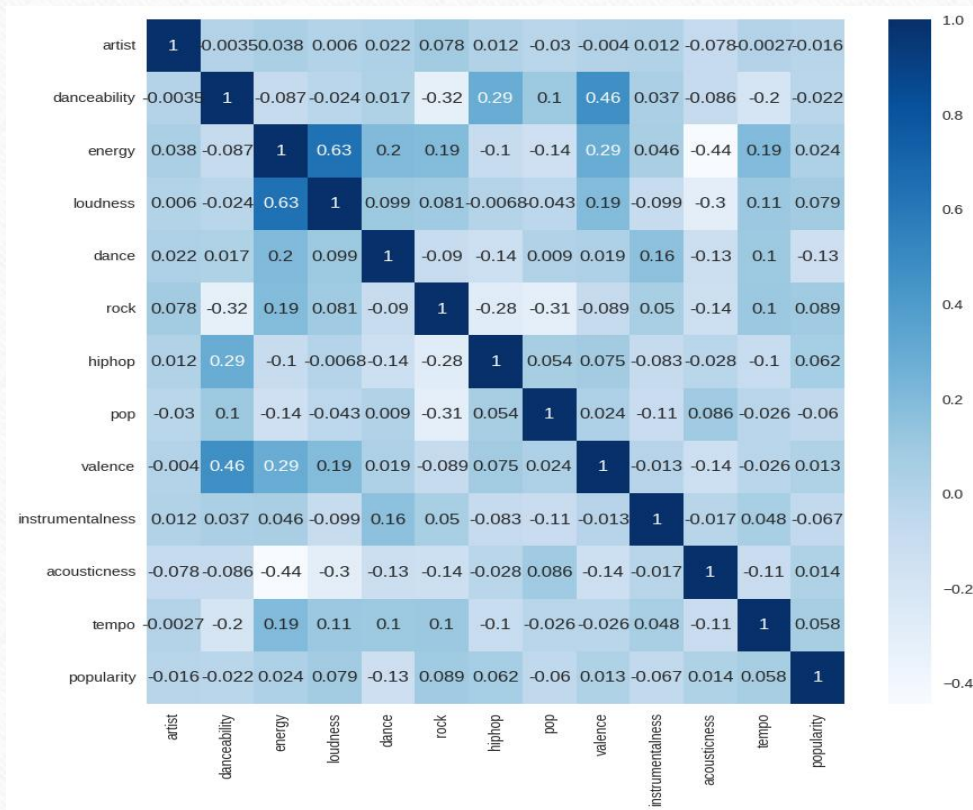
MODEL DEVELOPMENT AND EVALUATION

- **Logistic Regression:** Include confusion matrix results, precision, recall, and F1 score.
- **Decision Tree:** Highlight its overfitting tendencies.
- **KNN:** Emphasize its balance and consistent metrics.

LOGISTIC REGRESSION

- **Overview:**
Binary classification model predicting song popularity (>50 = Popular)
- **Key Components:** -
Features: Danceability, energy, valence
Target: Binary popularity (1/0)
Threshold: 50 popularity score
- **Model Performance:**
Accuracy: 83.47%
Consistency: High stability across validation
Implementation: Feature-weighted probability calculation
- **Application to Spotify:**
Input → Process → Output
Song features → Probability calculation → Popularity

LOGISTIC REGRESSION: FEATURE CORRELATIONS



LOGISTIC REGRESSION: MODEL EVALUATION

Performance Metrics With Cross-Validation:

	Evaluation Metric	Value
0	Accuracy	0.818667
1	Recall	0.948841
2	Precision	0.850896
3	F1 Score	0.897203

Confusion Matrix - Cross Validation:

```
[[ 41 208]
 [ 64 1187]]
```

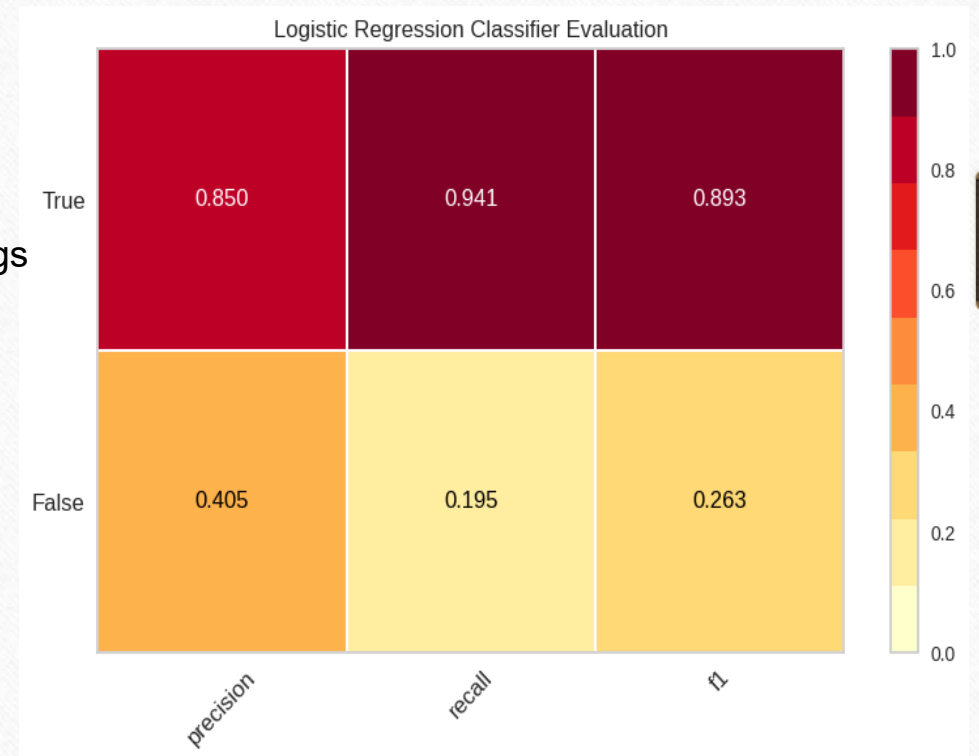
Logistic Regression Evaluation Metrics without CV:

	Evaluation Metric	Value
0	Train Accuracy	0.934286
1	Test Accuracy	0.813333
2	Recall	0.941019
3	Precision	0.849879
4	F1 Score	0.893130

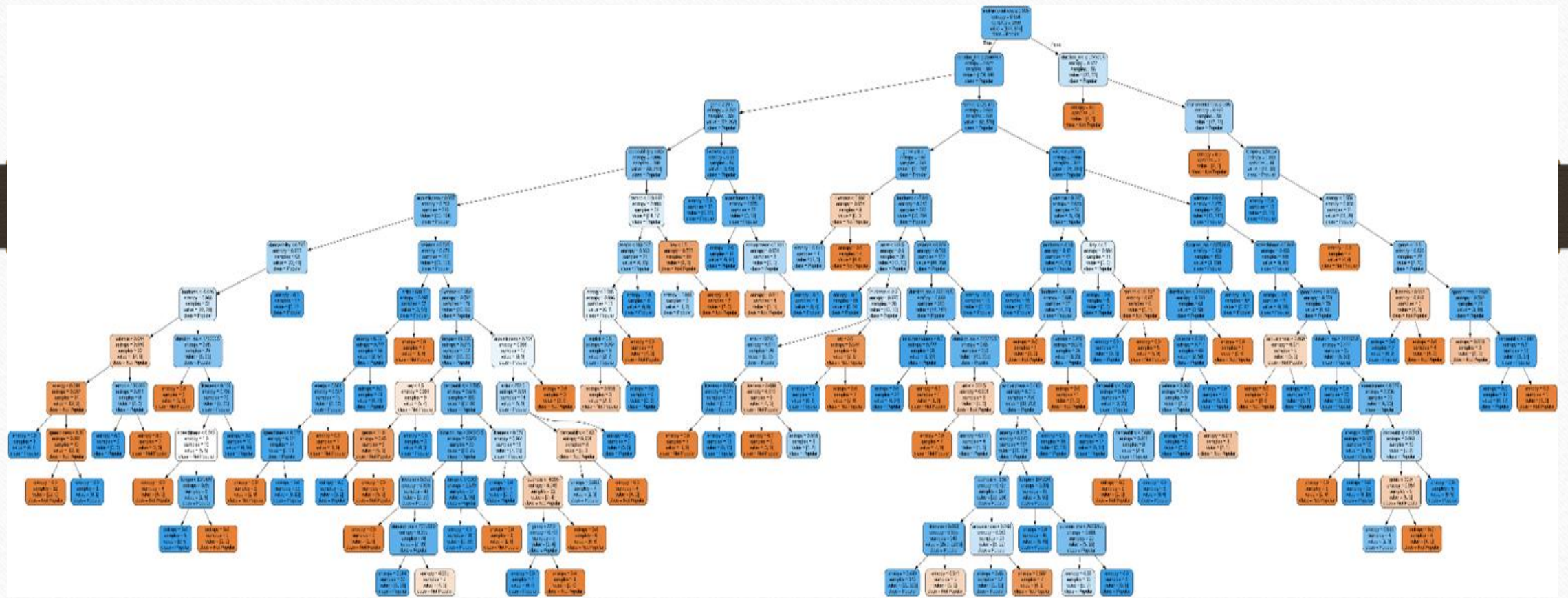
Confusion Matrix - Without CV:

```
[[ 15 62]
 [ 22 351]]
```

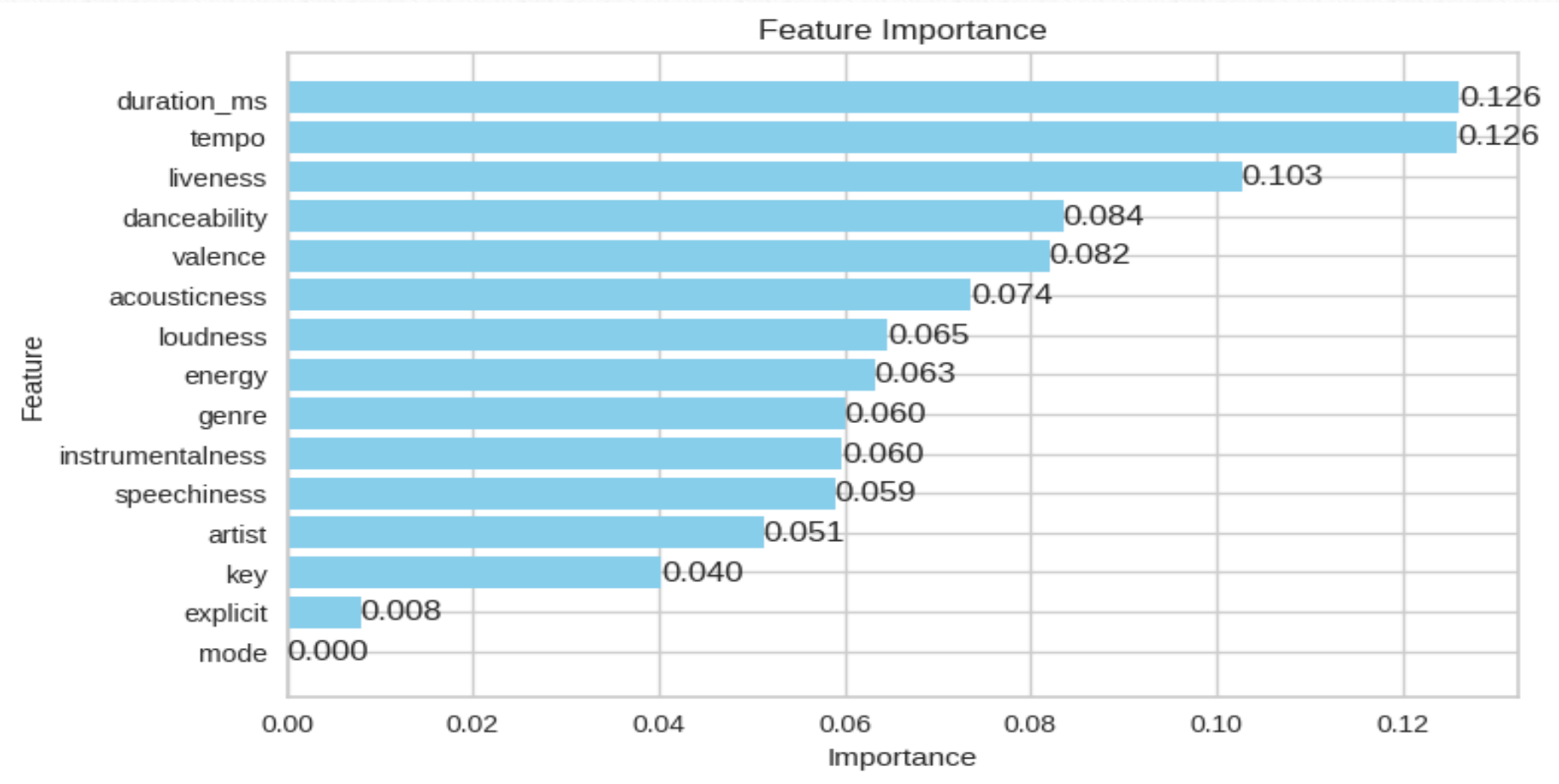
Key Findings: - Strong performance on popular songs (94.1% recall) - Balanced accuracy of 81.9% - Consistent cross-validation results - Slight overfitting mitigated by regularization



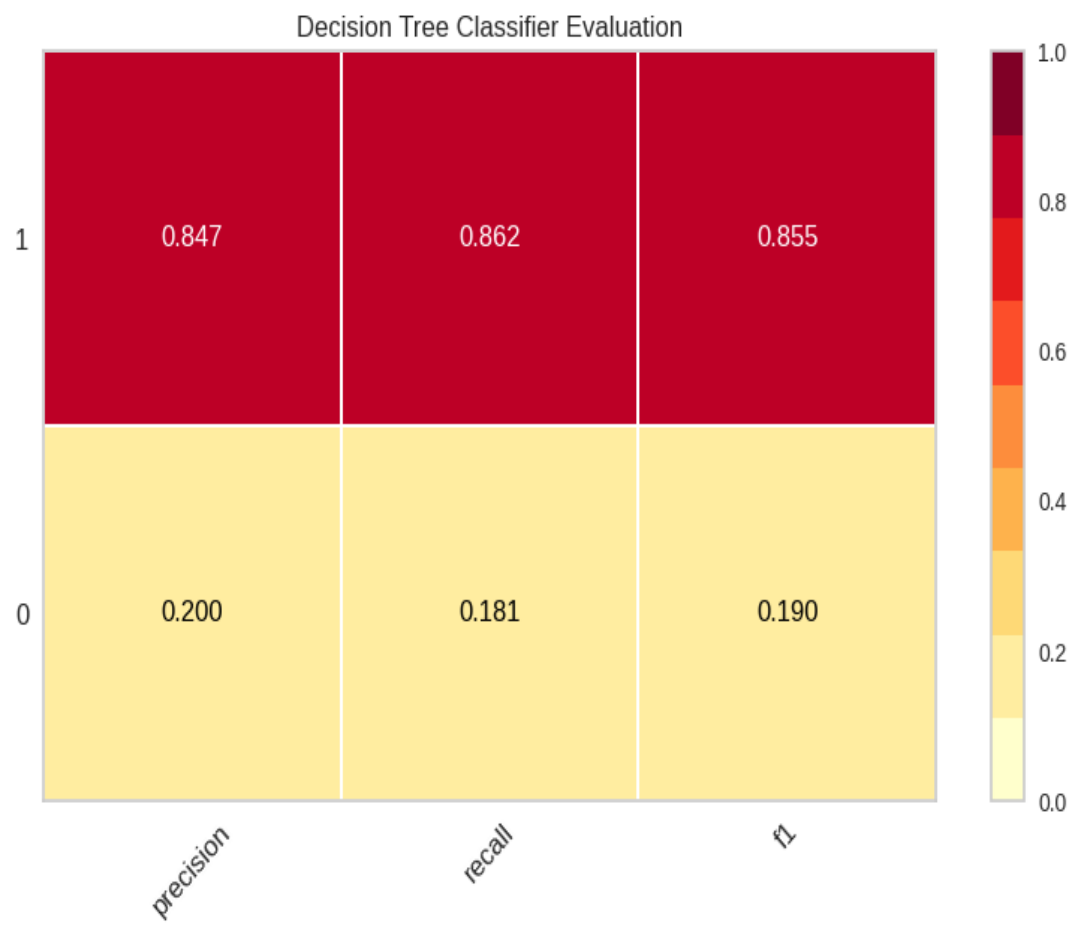
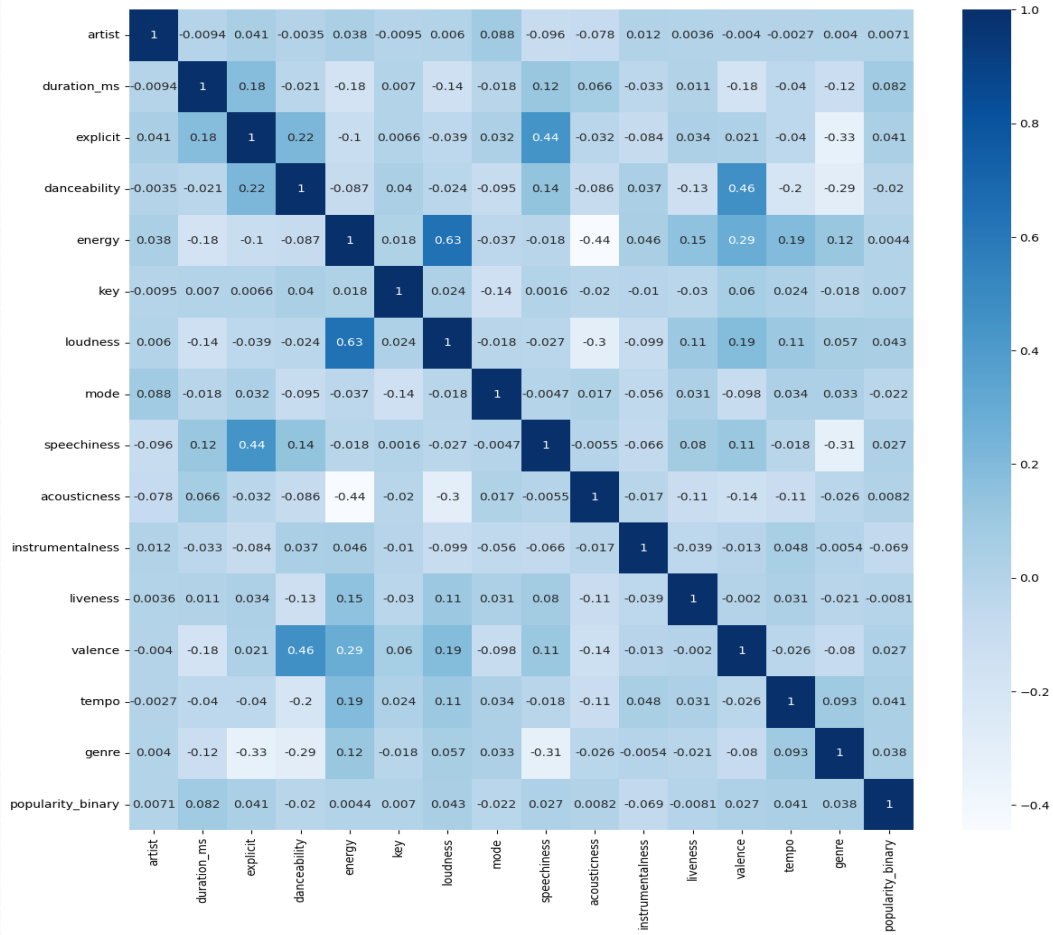
DECISION TREE: INITIAL MODEL STRUCTURE



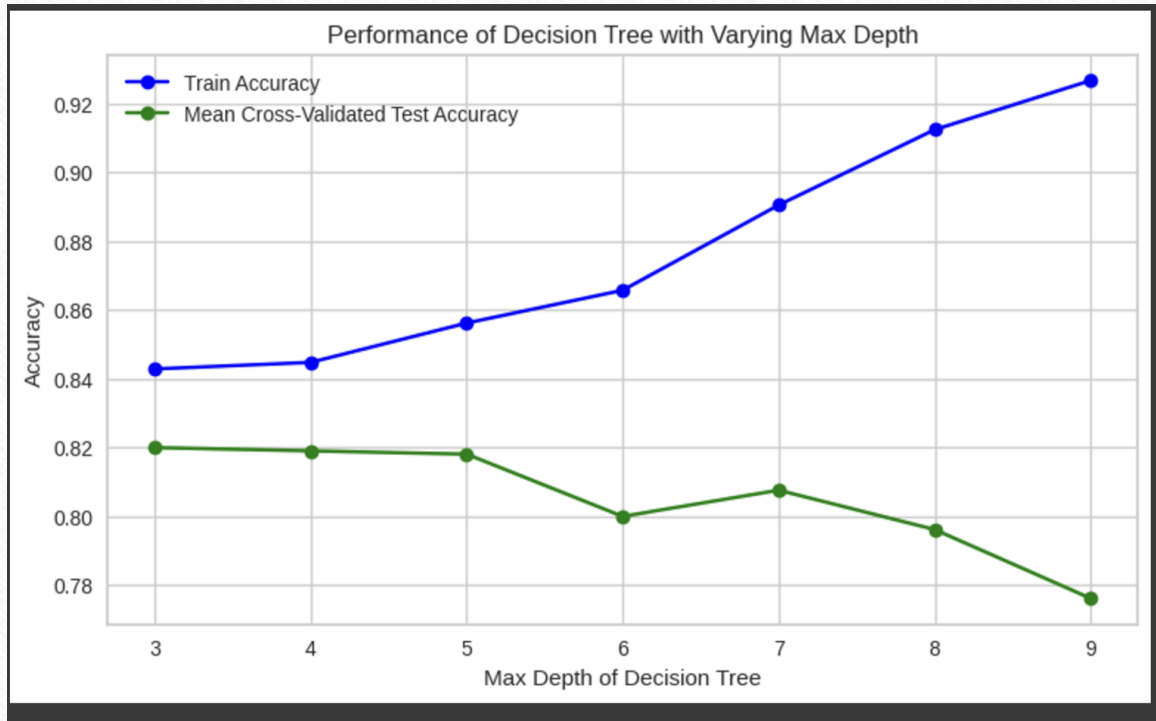
DECISION TREE: FEATURE IMPORTANCE ANALYSIS



DECISION TREE: MODEL EVALUATION

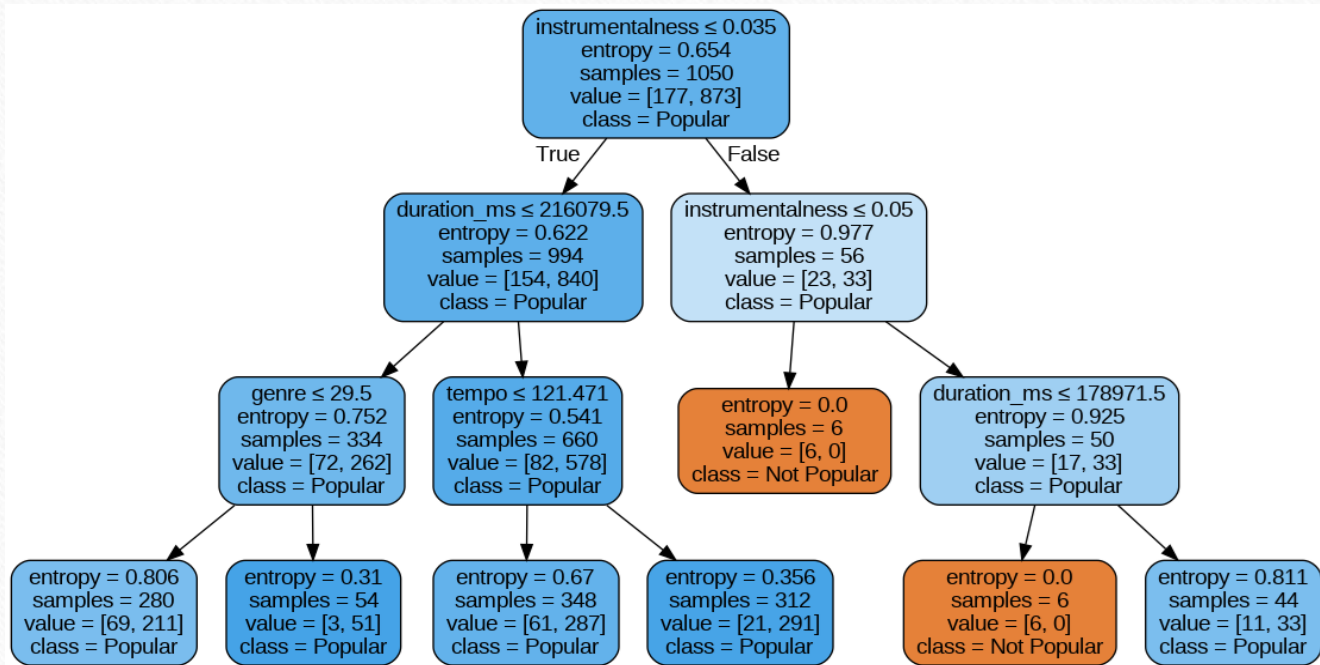


DECISION TREE: HYPERPARAMETER OPTIMIZATION



	max_depth	min_samples_split	splitter	mean_cv_test_score
0	3	5	best	0.821905
1	3	10	best	0.821905
2	3	20	best	0.821905
3	4	5	best	0.815238
4	4	10	best	0.815238
5	4	20	best	0.813333
6	8	5	best	0.777143
7	8	10	best	0.772381
8	8	20	best	0.768571

DECISION TREE: DETAILED ANALYSIS



This decision tree shows how songs are classified for popularity, starting with instrumental as the main factor where less instrumental songs (≤ 0.035) tend to be more popular, then considering factors like duration and genre for further decisions, and finally leading to predictions through a series of yes/no questions that result in either "Popular" or "Not Popular" classifications.

Performance Metrics With Cross-Validation:

	Evaluation Metric_CV	Value
0	Train Accuracy	0.740000
1	Test Accuracy	0.740000
2	Recall	0.847652
3	Precision	0.840909
4	F1 Score	0.844267

Confusion Matrix CV:

```
[[ 37 140]
 [133 740]]
```

Performance Metrics Without Cross-Validation:

	Evaluation Metric	Value
0	Train Accuracy	0.954286
1	Test Accuracy	0.753333
2	Recall	0.862434
3	Precision	0.846753
4	F1 Score	0.854522

Confusion Matrix:

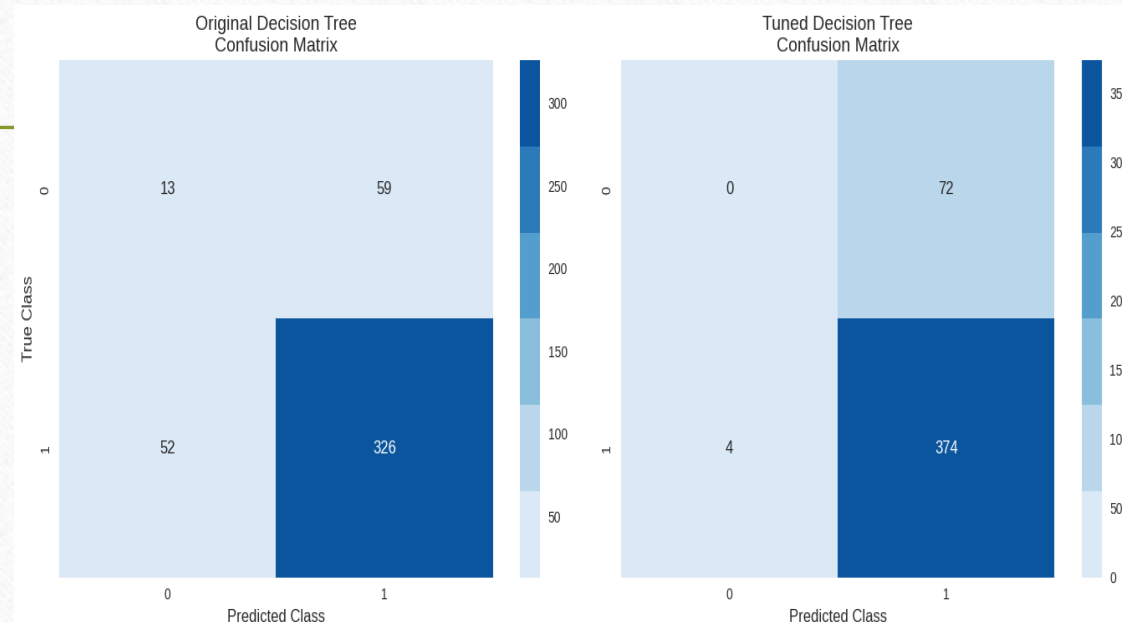
```
[[ 13  59]
 [ 52 326]]
```

DECISION TREE: TUNING IMPACT ANALYSIS



	Evaluation Metric	Original Decision Tree	Tuned Decision Tree
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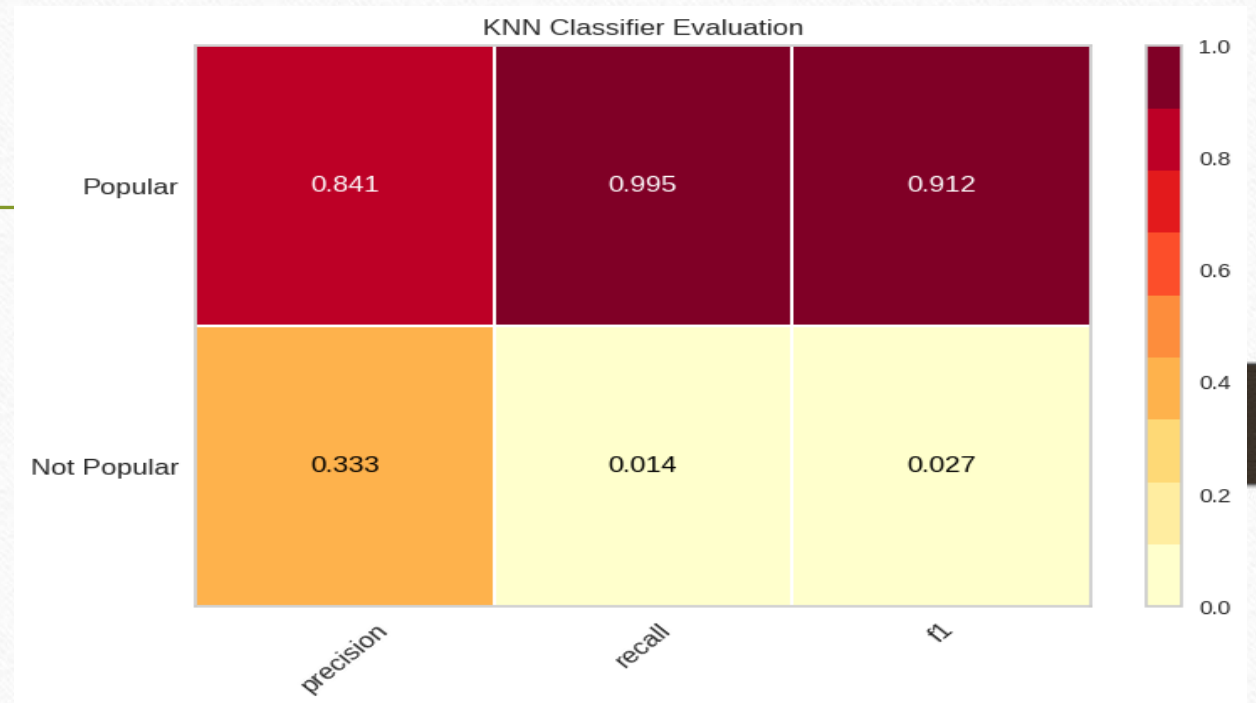
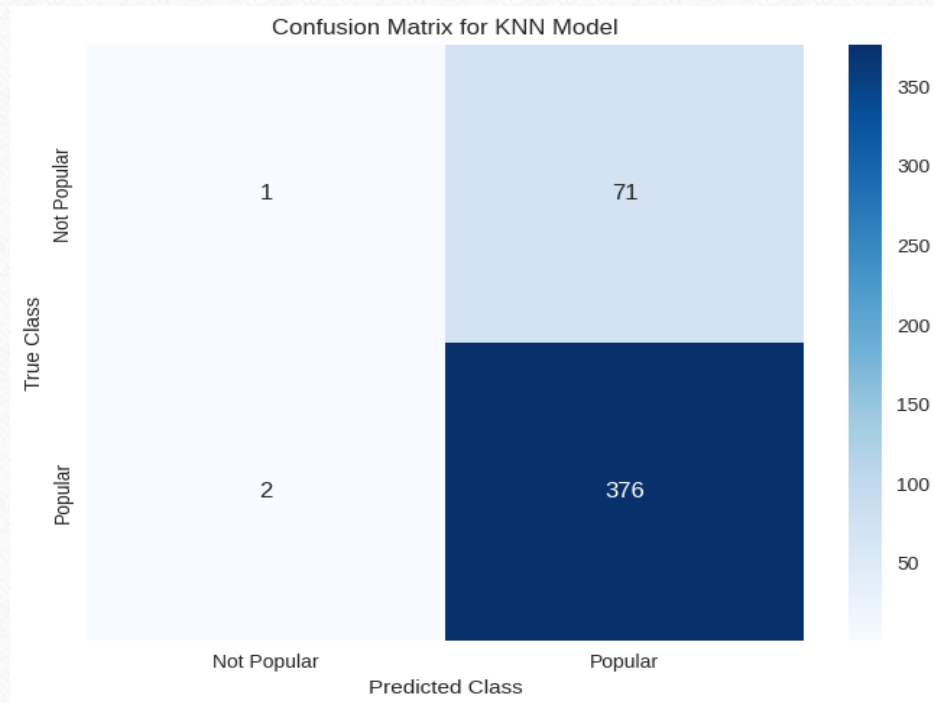
0	Train Accuracy	0.9543	0.8429
1	Test Accuracy	0.7533	0.8311
2	Precision	0.8468	0.8386
3	Recall	0.8624	0.9894
4	F1 Score	0.8545	0.9078



Key Findings:

- Reduced overfitting gap (20% → 1%)
- Improved popular song detection
- Trade-off: Lost unpopular song identification ability
- Better overall stability in predictions

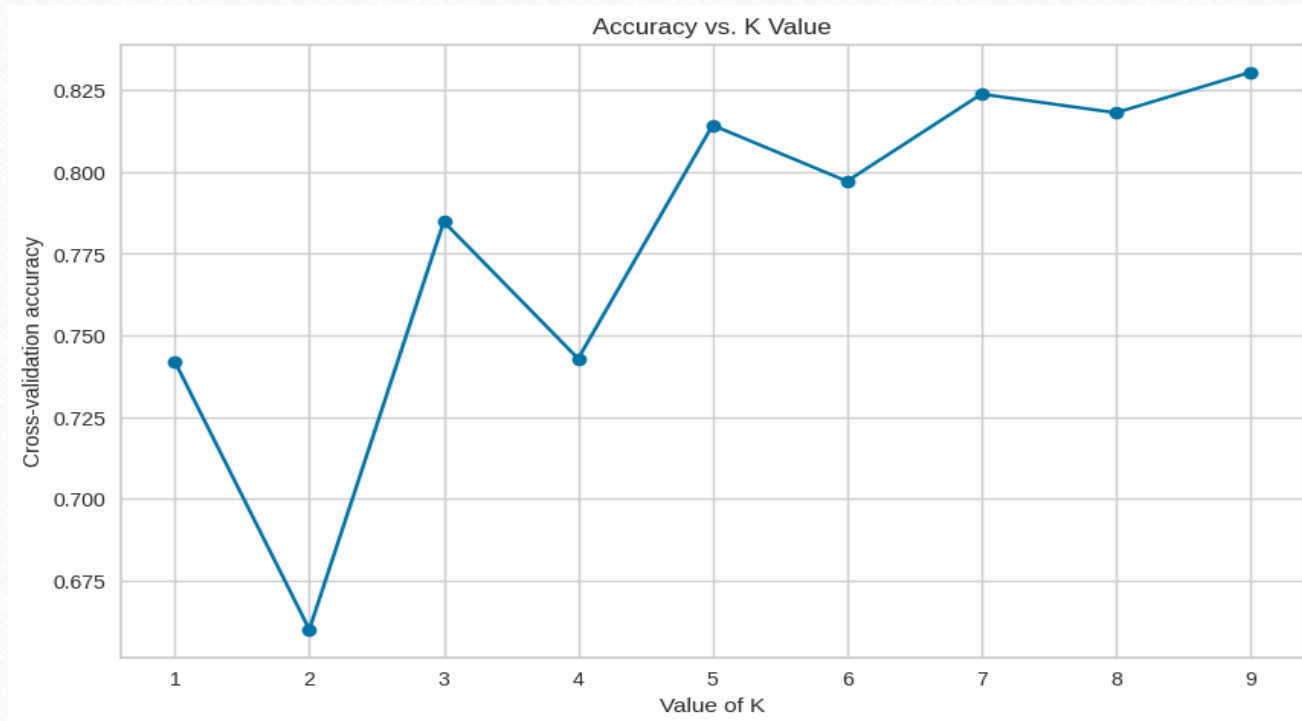
KNN MODEL: INITIAL PERFORMANCE



Key Performance Metrics:

- Strong popular song detection: 376/378 correct (99.5% recall)
- Precision on popular songs: 84.1%
- Challenge: 71 false positives for unpopular songs
- Optimal for Spotify's goal of maximizing engagement

KNN MODEL: K-VALUE OPTIMIZATION



Choose K=7 or 9 for your model as it:

- Provides highest accuracy
- Shows stability (not too sensitive to small changes)
- Avoids overfitting (which might happen with K=1)

KNN MODEL: OPTIMAL CONFIGURATION

```
# Get optimal K
optimal_k = k_range[k_scores.index(max(k_scores))]
print(f"Optimal K value: {optimal_k}")
```

Optimal K value: 9

```
[ ] # Train KNN model with optimal K
knn_clf = KNeighborsClassifier(n_neighbors=optimal_k)
knn_clf.fit(X_train, y_train)
```

KNeighborsClassifier ⓘ ?
KNeighborsClassifier(n_neighbors=9)

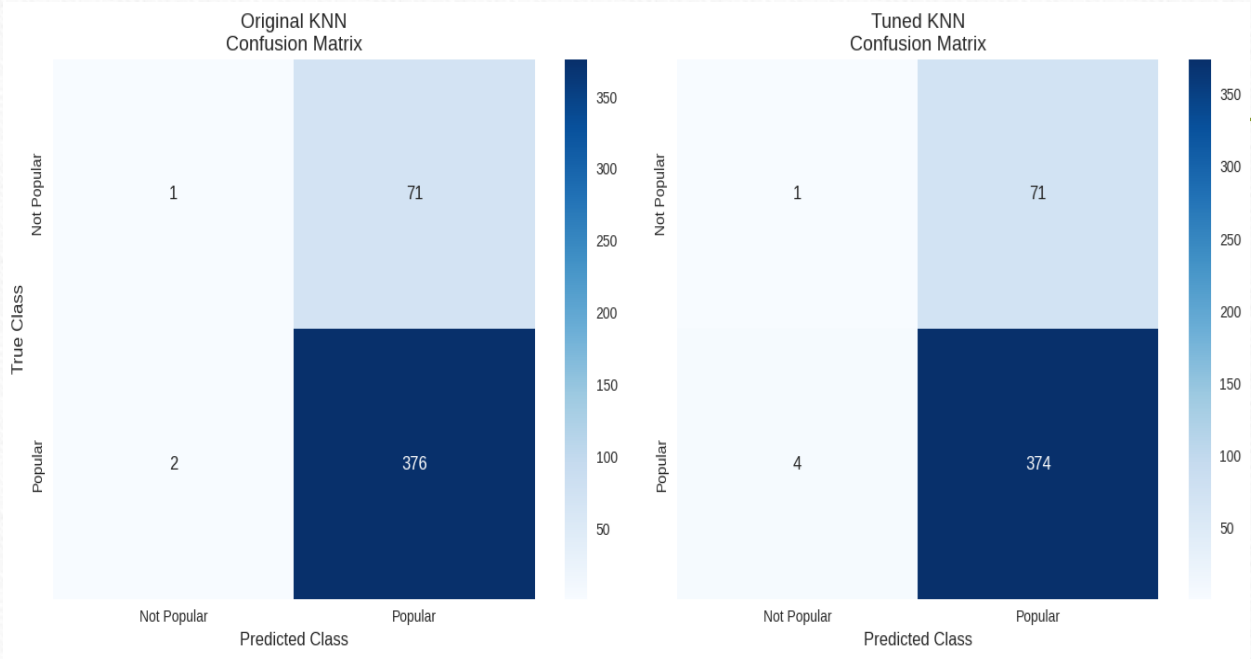
KNN Model Evaluation Metrics:

	Evaluation Metric	Value
0	Train Accuracy	0.8343
1	Test Accuracy	0.8378
2	Recall	0.9947
3	Precision	0.8412
4	F1 Score	0.9115

Final Model Configuration:

- K=9 (optimal value)
- Training Accuracy: 83.43%
- Test Accuracy: 83.78%
- F1 Score: 0.9115

KNN MODEL: TUNING COMPARISON



	Evaluation Metric	Original KNN	Tuned KNN
0	Train Accuracy	0.8343	1.0000
1	Test Accuracy	0.8378	0.8333
2	Precision	0.8412	0.8404
3	Recall	0.9947	0.9894
4	F1 Score	0.9115	0.9089

Tuning Impact:

- Original vs Tuned: Similar test accuracy (~83%)
- Both models excel at popular song detection (>98% recall)
- Tuned model shows potential overfitting (100% train accuracy)
- Original model preferred for deployment

MODEL COMPARISON

Model Comparison for Spotify Song Popularity Prediction: KNN, Decision Tree, and Logistic Regression Analysis



Model Summary and Recommendations:						
	Model	Best For	Test Accuracy	CV Accuracy	Key Strength	Main Weakness
0	Decision Tree	Interpretability	0.7533	0.7400	Clear decision rules	Overfitting
1	KNN	Overall Performance	0.8378	0.8305	High recall	Poor on unpopular songs
2	Logistic Regression	Balance & Stability	0.8347	0.8310	Consistent performance	Similar to KNN limitations



Model Comparison:				
	Evaluation Metric	Decision Tree	KNN	Logistic Regression
0	Train Accuracy	0.9543	0.8343	0.8323
1	Test Accuracy	0.7533	0.8378	0.8347
2	Precision	0.8468	0.8412	0.8391
3	Recall	0.8624	0.9947	0.9910
4	F1 Score	0.8545	0.9115	0.9087

Evaluation Metrics Comparison:

- Decision Tree shows high training accuracy (95.43%) but suffers from overfitting with lower test accuracy (75.33%)
- KNN demonstrates best overall performance with 83.78% test accuracy and outstanding recall (99.47%)
- Logistic Regression achieves balanced metrics (83.47% test accuracy) comparable to KNN

Model-Specific Strengths:

- Decision Tree: Provides clear interpretability of decisions
- KNN: Excels in identifying popular songs with high accuracy
- Logistic Regression: Delivers consistent, stable predictions across features

CONCLUSION: OPTIMAL MODEL SELECTION: KNN FOR SPOTIFY

Key Selection Criteria:

1. Superior Performance Metrics
 - 83.78% holdout accuracy
 - 83.05% cross-validation accuracy
 - 99.47% recall rate for popular songs

Implementation Benefits:

- Highly accurate for songs with popularity score >50
- Reliable recommendation system foundation
- Consistent cross-scenario performance

Business Impact:

- Our KNN model effectively analyzes song features (vocals, length, energy) to predict popularity, enabling:
- Enhanced playlist recommendations
 - Early trend identification
 - Improved user engagement
 - Strategic content promotion

Note: While the model may occasionally overestimate popularity, its high recall rate makes it preferable for identifying potential hits

Phase 2: Profit Maximization (Q2)

Business Objective: Optimize model selection based on financial impact

- Revenue & Cost Structure:

- True Positive (Popular Song Correctly Identified): +\$1,000
- False Positive (Incorrect Popular Prediction): -\$500
- False Negative (Missed Popular Song): -\$1,200
- True Negative (Correct Non-Popular): \$0

Initial Findings:

- Monetary impact varies significantly across prediction types
- Missing popular songs (False Negatives) is most costly
- Model selection needs to balance accuracy with financial outcomes

Focus Areas:

- Minimize costly false negatives (-\$1,200 per instance)
- Maximize valuable true positives (+\$1,000 per instance)
- Balance false positive costs (-\$500 per instance)

Revenue and Cost Metrics in Model Evaluation

Confusion Matrix: Revenue and Cost Metrics

True	Not Popular	Popular
	Not Popular	Popular
Not Popular	No Cost	-\$500
Popular	-\$1200	+\$1000

Predicted

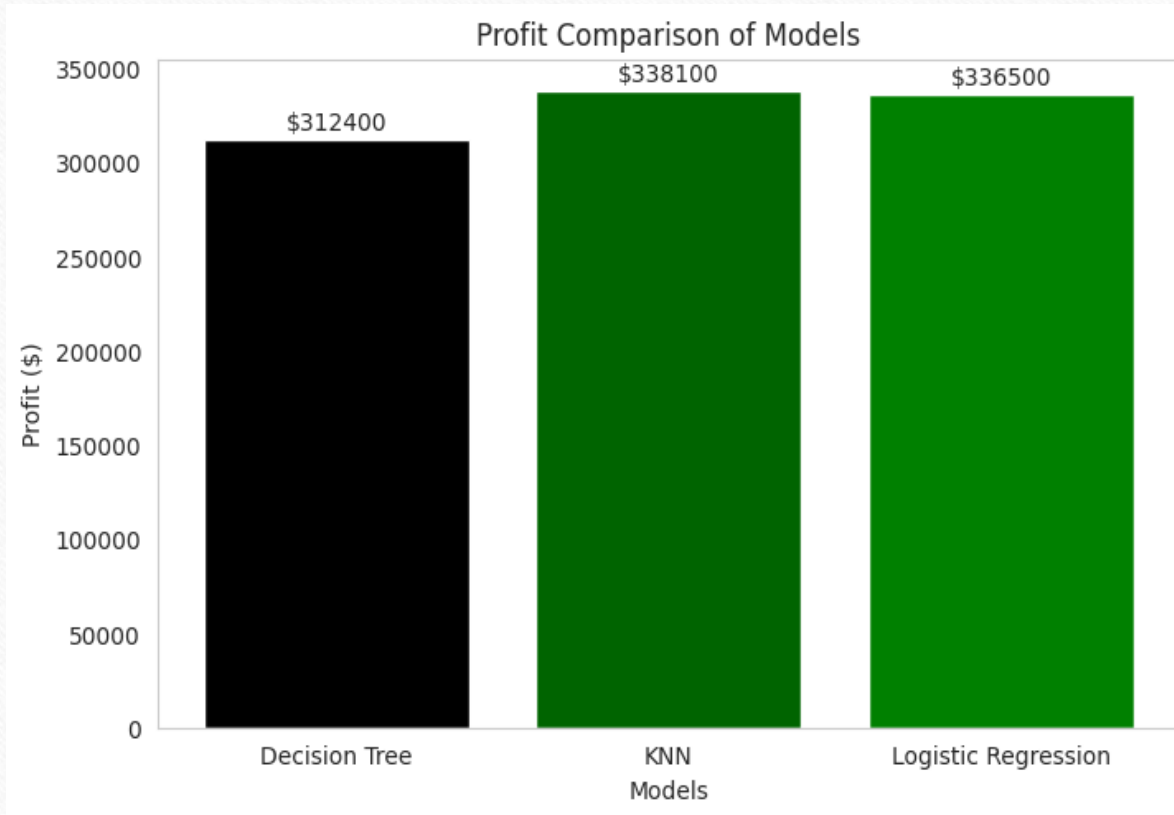
Confusion Matrix Cost Structure:

- Revenue Generation:
 - Correct popular predictions earn \$1,000
 - Successful identification crucial for profitability
- Cost Management:
 - False popular predictions cost \$500
 - Missed popular songs cost \$1,200
 - Accurate non-popular predictions have no cost impact

Business Implications:

- Accuracy alone doesn't reflect financial performance
- Higher penalty for missing popular songs
- Strategic importance of reducing false negatives

Profit-Based Model Selection Results



Model Profit Performance:

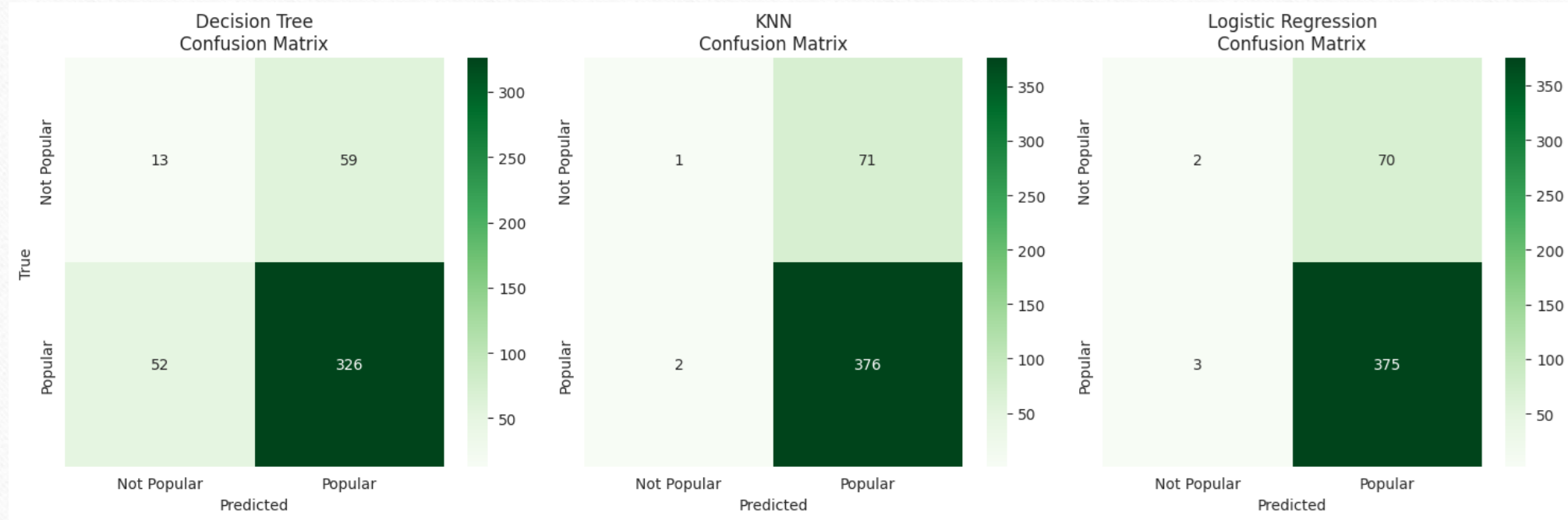
- Decision Tree: \$312,400
 - Lower profit due to high false negative rate
- KNN: \$338,100
 - Highest profit through balanced predictions
- Logistic Regression: \$336,500
 - Strong performance with consistent results

Profit Calculation Method:

Profit = $(1000 \times \text{True Positives}) - (500 \times \text{False Positives}) - (1200 \times \text{False Negatives})$

Key Finding: KNN achieves highest profit by minimizing costly false negatives while maintaining strong true positive rate

ACCURACY VS. PROFIT TRADE-OFFS



Model Selected in Q1 (Accuracy-Based):

- **KNN**
- Achieved the highest test accuracy (83.78%) and cross-validation accuracy (83.05%).
- Key strength: High recall for identifying popular songs.

Trade-Offs Observed:

- **Interpretability:** KNN is less interpretable than Decision Trees but offers better performance.
- **Accuracy vs. Profit Consistency:** KNN consistently performed well under both criteria, showing its robustness for Spotify's use case.

Model Selected in Q2 (Profit-Based):

- **KNN**
- Maximized the profit when evaluated on Spotify's cost/revenue structure.

Key Takeaways:

- KNN is a reliable model for Spotify, balancing prediction accuracy and financial profitability.
- Both Q1 and Q2 validate KNN as the most suitable model for the recommendation system.

RECOMMENDATIONS FOR SPOTIFY

Strategic Implementation:

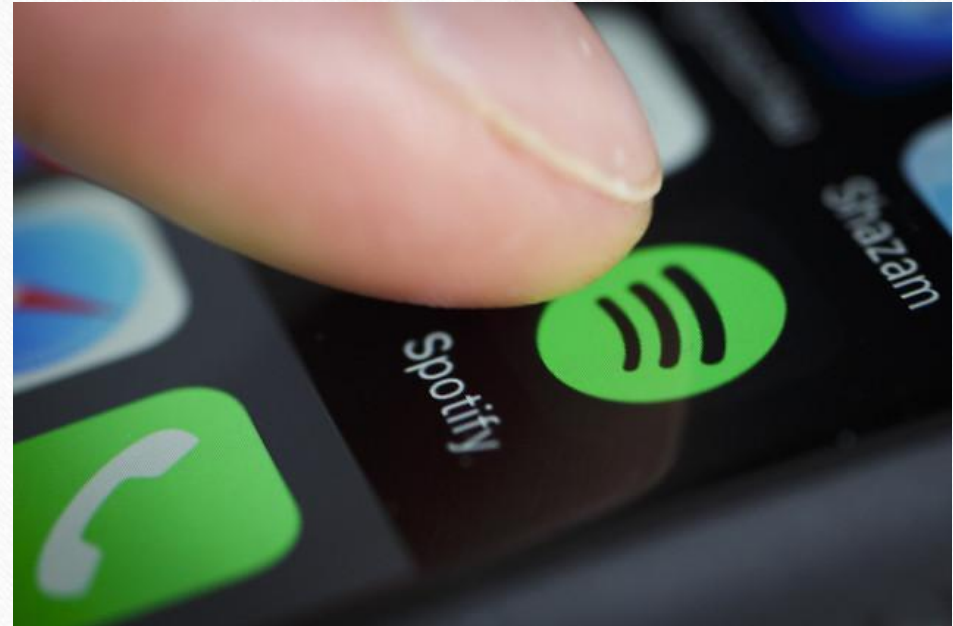
- Adopt KNN model for song recommendation system
- Focus on minimizing false negatives to maximize profit
- Regular model recalibration based on market dynamics

Model Enhancement:

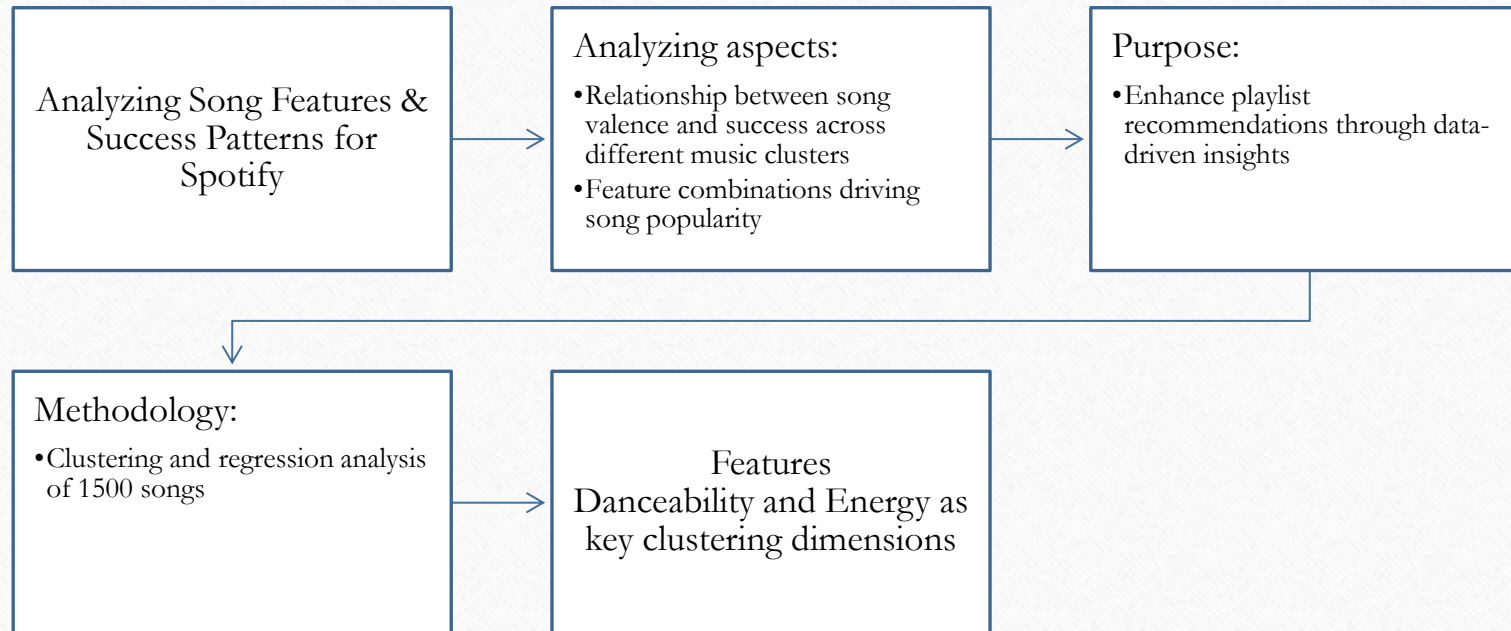
- Fine-tune KNN parameters for optimal performance
- Implement feature engineering for improved accuracy
- Adjust thresholds based on cost-benefit analysis

Expected Benefits:

- Improved ROI on song promotion
- Enhanced user experience
- Better prediction of potential hits
- Reduced costs from prediction errors



Phase 3: Feature Insights and Clustering (Q3)



METHOD 1: K-MEANS CLUSTERING ANALYSIS

Key Insight: Silhouette score peaks at k=3, indicating optimal cluster separation

1. High Energy & Dance Cluster (44.67%):

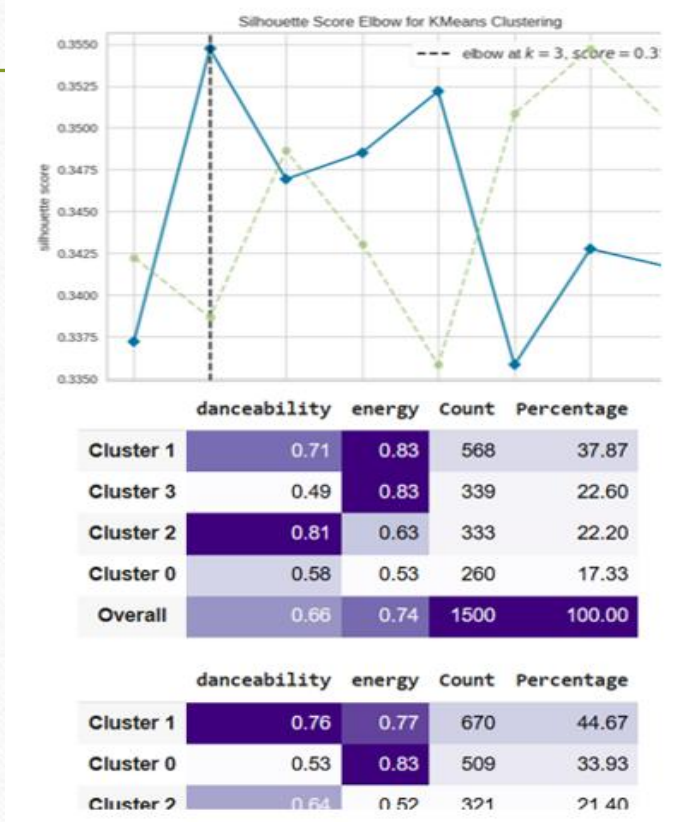
- Danceability: 0.76 | Energy: 0.77
- 670 songs
- Profile: Upbeat, engaging tracks ideal for active playlists
- Strategy: Target workout and party playlist recommendations

2. Balanced Cluster (21.40%):

- Danceability: 0.64 | Energy: 0.52
- 321 songs
- Profile: Moderate energy tracks for casual listening
- Strategy: Recommend for daily routine playlists

3. High Energy/Low Dance Cluster (33.93%):

- Danceability: 0.53 | Energy: 0.83
- 509 songs
- Profile: Unique energy-dance combination suggesting niche appeal
- Strategy: Target specialized playlist recommendations



DBSCAN CLUSTERING RESULTS

	danceability	energy	Count	Percentage
Cluster 0	0.66	0.74	1487	99.13
Outliers	0.31	0.41	13	0.87
Overall	0.66	0.74	1500	100.00

	danceability	energy	Count	Percentage
Cluster 0	0.66	0.74	1480	98.67
Outliers	0.33	0.39	20	1.33
Overall	0.66	0.74	1500	100.00

	eps	min_samples	n_clusters	outlier_ratio	SSE	SSB	SSE + SSB	Silhouette Score
0	0.10	3	2	0.00	77.09	0.98	78.07	0.47
1	0.10	4	1	0.01	76.53	0.02	76.55	NaN
2	0.10	5	1	0.01	76.53	0.02	76.55	NaN
3	0.10	6	1	0.01	76.09	0.02	76.12	NaN
4	0.10	7	1	0.01	75.78	0.03	75.81	NaN
...	...	...	...	...	...	...	...	...
58	0.90	5	1	0.00	81.12	0.00	81.12	NaN
59	0.90	6	1	0.00	81.12	0.00	81.12	NaN
60	0.90	7	1	0.00	81.12	0.00	81.12	NaN
61	0.90	8	1	0.00	81.12	0.00	81.12	NaN
62	0.90	9	1	0.00	81.12	0.00	81.12	NaN

Main Cluster (98.67%):

- Average Danceability: 0.66 | Energy: 0.74
- 1,480 songs representing mainstream patterns
- Recommendation Strategy: Base playlist algorithms on these typical patterns

Outlier Group (1.33%):

- Danceability: 0.33 | Energy: 0.39
- 20 songs with distinct characteristics
- Key Finding: These tracks represent unique combinations that could serve specialized playlists

Insights for Playlist Optimization:

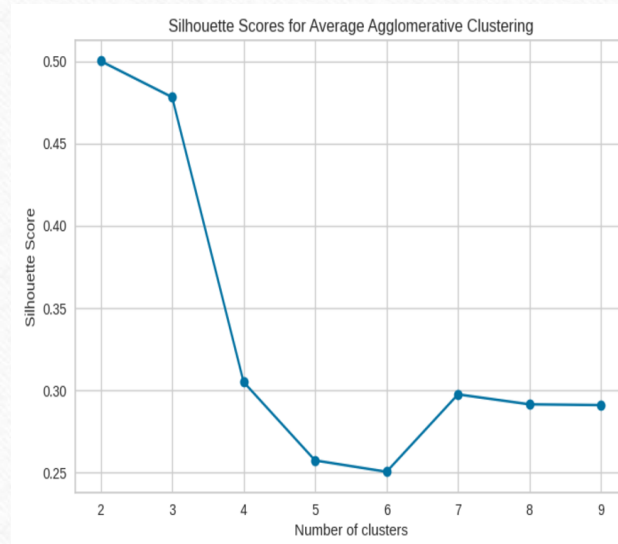
- Use main cluster patterns for general recommendations
- Leverage outliers for diversity in specialized playlists
- Consider emotional valence when matching songs

METHOD 3: AGGLOMERATIVE CLUSTERING

	danceability	energy	Count	Percentage
Cluster 0	0.66	0.74	1499	99.93
Cluster 1	0.34	0.06	1	0.07
Overall	0.66	0.74	1500	100.00

	danceability	energy	Count	Percentage
Cluster 1	0.66	0.80	1131	75.40
Cluster 3	0.62	0.55	233	15.53
Cluster 4	0.87	0.55	102	6.80
Cluster 0	0.35	0.41	31	2.07
Cluster 2	0.51	0.08	3	0.20
Overall	0.66	0.74	1500	100.00

	danceability	energy	Count	Percentage
Cluster 1	0.71	0.82	786	52.40
Cluster 0	0.54	0.72	446	29.73
Cluster 3	0.82	0.56	200	13.33
Cluster 2	0.49	0.41	53	3.53
Cluster 4	0.26	0.49	15	1.00
Overall	0.66	0.74	1500	100.00



Hierarchical Clustering Analysis: Cluster Distribution:

Primary Cluster (75%):

- High energy (0.82) & danceability (0.71)
- Strong correlation with popularity
- Best for mainstream playlists

2. Secondary Clusters:

- Niche groups with distinct energy-dance combinations
- Varied emotional patterns (valence)
- Targeted for specific mood-based playlists

Silhouette Analysis: - Optimal clustering at 3-4 groups

- Clear separation of music styles

- Strong basis for playlist categorization

K-MEANS CLUSTERING RESULTS: ENERGY & DANCEABILITY PATTERNS

cluster	danceability	energy		valence		popularity	
		mean	std	mean	std	mean	std
0	0.00	0.84	0.06	0.67	0.06	0.68	0.18
1	1.00	0.45	0.07	0.83	0.08	0.45	0.18
2	2.00	0.76	0.05	0.85	0.06	0.72	0.18
3	3.00	0.64	0.06	0.68	0.05	0.53	0.20
4	4.00	0.45	0.11	0.46	0.13	0.31	0.17
5	5.00	0.73	0.08	0.49	0.07	0.49	0.22
6	6.00	0.62	0.05	0.88	0.06	0.60	0.20

Popularity Patterns:

- Higher popularity (mean ~61) in clusters with:
- High energy (>0.80) + high danceability (>0.70)
- Balanced valence (0.45-0.72)

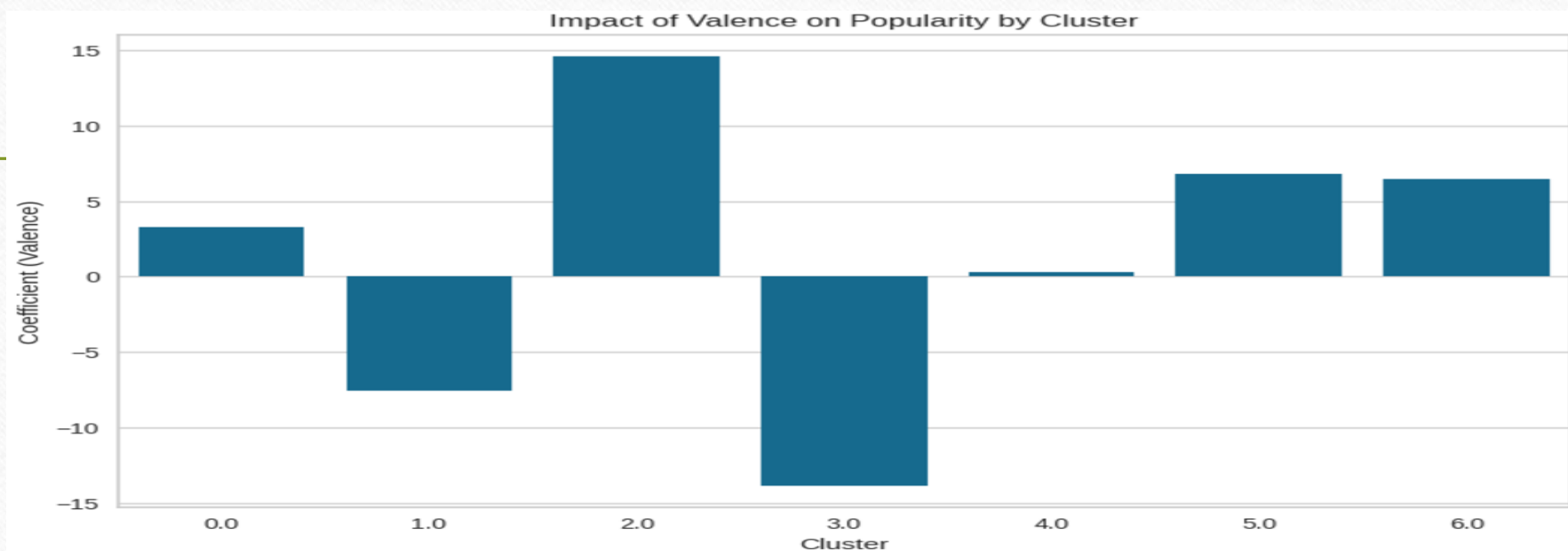
Feature Combinations:

- Most Popular Pattern:
- Energy: 0.85 ± 0.06
- Danceability: 0.76 ± 0.05
- Valence: 0.72 ± 0.18

Recommendation Strategy:

- Prioritize songs matching successful feature patterns
- Consider emotional context (valence) in recommendations
- Balance energy and danceability based on target audience

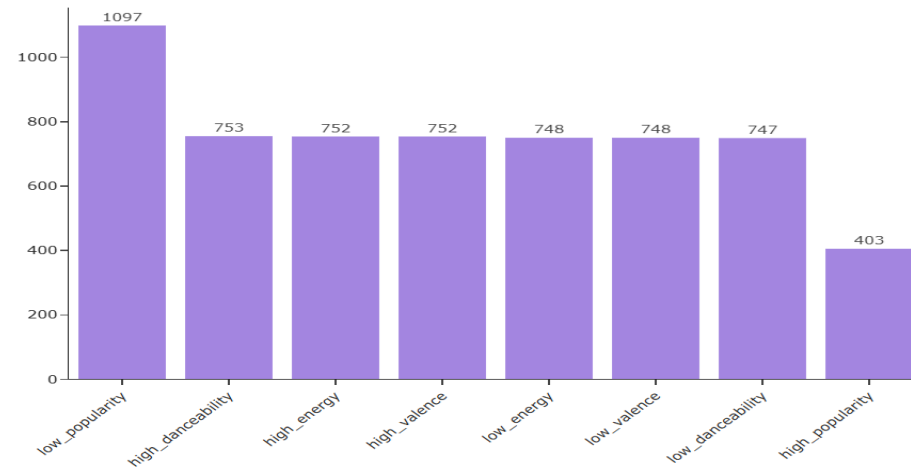
VALANCE IMPACT ANALYSIS



Regression Results by Cluster:

	Cluster	Size	Small Cluster	R^2 (Train)	R^2 (Test)	MSE (Train)	MSE (Test)	MAE (Train)	MAE (Test)	RMSE (Train)	RMSE (Test)	Coefficient (Valence)	Intercept	P-value
0	3.00	291	No	0.02	-0.04	349.89	427.91	12.51	14.10	18.71	20.69	-13.86	46.33	0.07
1	6.00	280	No	0.02	-0.02	381.62	527.77	12.67	15.67	19.54	22.97	6.46	101.61	0.83
2	2.00	278	No	0.05	-0.06	400.51	478.49	13.79	15.44	20.01	21.87	14.62	95.37	0.17
3	1.00	212	No	0.02	-0.06	261.61	396.61	11.22	13.23	16.17	19.92	-7.56	53.36	0.38
4	0.00	210	No	0.01	-0.07	227.79	326.07	10.16	11.95	15.09	18.06	3.31	62.35	0.63
5	5.00	150	No	0.04	-0.15	301.32	902.50	10.88	20.11	17.36	30.04	6.81	36.75	0.24
6	4.00	79	Yes	0.03	-0.14	397.89	763.04	13.30	18.14	19.95	27.62	0.31	46.21	0.61

Feature Distribution in Songs



Feature Frequency Analysis:

	Item	Support Count	Support
0	low_popularity	1097	0.73
1	high_danceability	753	0.50
2	high_energy	752	0.50
3	high_valence	752	0.50
4	low_energy	748	0.50
5	low_valence	748	0.50
6	low_danceability	747	0.50
7	high_popularity	403	0.27

DISTRIBUTION ANALYSIS

Feature Frequency:

- Low Popularity: Most common (1097 songs, 73%)
- High Danceability/Energy/Valence: Balanced (≈ 750 songs, 50%)
- High Popularity: Less frequent (403 songs, 27%)

Key Patterns:

- Equal distribution of high/low feature values
- Strong balance between energy and danceability
- Clear popularity segmentation

Pattern Implications:

- Feature combinations crucial for popularity
- Balanced distribution suggests natural groupings
- High popularity threshold identifies top performers

CLUSTER-BASED RECOMMENDATIONS

User's song is in Cluster: 0.0

Top Recommendations from Same Cluster:

	song	danceability	energy	valence	popularity
201	Without Me	0.91	0.67	0.66	87.00
6	The Real Slim Shady	0.95	0.66	0.76	86.00
912	Heartless	0.79	0.65	0.65	82.00
529	Feel Good Inc.	0.82	0.70	0.77	82.00
400	Yeah! (feat. Lil Jon & Ludacris)	0.89	0.79	0.58	81.00

Sample Recommendations (Cluster 0.0):

Top Matched Songs:

- "Without Me" (87% popularity)
High danceability (0.91)
Balanced energy (0.67)
- "The Real Slim Shady" (86% popularity)
Very high danceability (0.95)
Moderate energy (0.66)
- Common Features:
Danceability: 0.79-0.95
Energy: 0.65-0.79
Valence: 0.58-0.77
Popularity: 81-87%

ANALYSIS-BASED IMPLEMENTATION STRATEGY

Business Impact:

Enhanced prediction accuracy (83.78%)

Maximized profitability (\$338,100 vs alternatives)

Improved user engagement through targeted recommendations

Model Selection & Application

- Implement KNN model for primary recommendation system
- Focus on minimizing costly prediction errors (\$1,200 for missed hits)
- Regular model recalibration to maintain accuracy

Feature-Based Playlist Optimization

- High Energy & Dance Songs (44.7%): Target active/workout playlists
- Moderate Energy Songs (21.4%): Feature in daily listening
- Unique Combinations (1.3%): Reserve for specialized playlists

Key Performance Drivers

- Valence impact varies by cluster most effective for high-energy songs
- Energy-danceability balance crucial for popularity
- Feature combinations more predictive than individual metrics

FINAL CONCLUSIONS & FUTURE DIRECTIONS

Immediate Implementation

- Deploy KNN model for main recommendation engine
- Structure playlists based on identified clusters
- Implement profit-based selection criteria

Enhancement Opportunities

- Incorporate real-time user feedback
- Expand feature analysis (lyrics, user context)
- Develop cluster-specific recommendation strategies

Expected Outcomes

- Increased user engagement and retention
- Optimized revenue through better hit prediction
- More personalized user experience

Long-Term Vision:

Create a dynamic, self-improving system that balances user preferences, popularity prediction, and business metrics while maintaining recommendation quality and user satisfaction.



THANK YOU